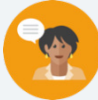


11.8 Triangle Midsegment Theorem

Lesson Introduction

 Benchmark Focus	MA.912.GR.1.3 Prove relationships and theorems about triangles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of triangles.		 Learning Targets	- I can discover the triangle midsegment theorem using AA similarity. - I can use the triangle midsegment theorem to prove relationships and to solve problems.	
 Benchmark Coherence	Building On			Working Toward	
	MA.912.GR.1.2 Prove triangle congruence or similarity using Side-Side-Side, Side-Angle-Side, Angle-Side-Angle, Angle-Angle-Side, Angle-Angle, and Hypotenuse-Leg.			MA.912.GR.1.4 Prove relationships and theorems about parallelograms. Solve mathematical and real-world problems involving postulates, relationships, and theorems of parallelograms.	
 Essential Questions	- What is the relationship between the triangle midsegment theorem and AA similarity? - How can you use the triangle midsegment theorem to solve problems?				
 Lesson Narrative	In this lesson, students use angle-angle similarity to discover the triangle midsegment theorem. Then, students use the triangle midsegment theorem to prove relationships and solve problems.				
 Component Summary	11.8.1 Warm-Up (~5 min.)	Notice and Wonder about a sculpture of cubes (<i>SE p. 220</i>)	11.8.3 Guided Practice (10-15 min.)	Use triangle midsegment theorem to solve guided problems (<i>SE p. 222</i>)	
	11.8.2 Exploration (20 min.)	Discover the triangle midsegment theorem (<i>SE p. 221</i>)	11.8.4 Your Turn! (5-10 min.)	Use triangle midsegment theorem to independently solve problems (<i>SE p. 224</i>)	
	11.8.5 Wrap-Up (~5 min.)	Identify and correct errors in a sample of student work (<i>SE p. 224</i>)			
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> Triangle Midsegment Theorem <u>Additional Glossary:</u> N/A		(Optional) Colored Pencils		N/A

 Teacher Tips	Common Misconceptions • Students may struggle with identifying similar triangles by Angle-Angle Similarity. • Students may struggle with using the triangle midsegment theorem and may miscalculate when using it, resulting in an incorrect answer.	Things to Consider • Consider providing colored pencils or highlighters for students to color code and label their diagrams throughout this lesson. • Consider using a dynamic tool, such as GeoGebra, to enable students to investigate the location of the midsegment.
	Literacy and Language Routines Notice and Wonder In 11.8.1 Warm-Up, students are shown a photograph of a sculpture made of cubes. Ask students “What do you notice? What do you wonder?” Give students a few minutes to write down things they notice, and things they wonder. After students have had a chance to write down their responses, ask several students to share things they noticed and wondered. Record these for all to see. Steer the conversation to wondering about how the sculpture remains balanced.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.4.1: Discussion and Reflection In 11.8.5 Wrap-Up, students have the opportunity to demonstrate their mastery of the learning targets by identifying and correcting errors in a sample of student work. Discussing and reflecting on the sample student errors will decrease the likelihood of students making similar errors themselves in the future.
	 Differentiate Instruction	Provide an auditory entry point in 11.8.2 Exploration by having students discuss what they notice about the relationship between the triangles. Provide a kinesthetic entry point by having students use Geoboards, Magformers, Busy Bags, or Exploragons to physically model the diagrams in 11.8.3 Guided Practice. Provide a visual entry point by prompting students to color code and label their diagrams in 11.8.4 Your Turn!
Lesson Notes:		

11.8.1 Warm-Up: “Many Small Cubes”

Component Overview: Students share what they notice and wonder about a sculpture of cubes. This task is intended to inspire student curiosity and interest. This component can help build students’ agency and mathematical identities, as each and every student can participate.



11.8.1 Warm-Up: “Many Small Cubes”

1. Japanese architect Sou Fujimoto exhibited a sculpture of stacked and suspended aluminum cubes as a part of the art fair in the Parisian Jardins des Tuileries’ gardens.



- a. What do you notice about the structure?
Answers will vary. I notice that there are all kinds of different sized cubes. I notice that there are gaps between the cubes.
- b. What do you wonder?
Answers will vary. I wonder how they all stay stacked the way they are. I wonder if there is anything holding them together.



Notice and Wonder

Students are shown a photograph of a sculpture made of cubes. Ask students “What do you notice? What do you wonder?” Give students a few minutes to write down things they notice, and things they wonder. After students have had a chance to write down their responses, ask several students to share things they noticed and wondered. Record these for all to see. Steer the conversation to wondering about how the sculpture remains balanced.

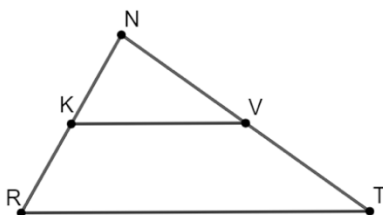
11.8.2 Exploration: Triangle Midsegment Theorem

Component Overview: Students use the fact that triangles have two congruent angles to prove that the triangles are similar. Then, students use the proportionality of the side lengths to determine missing side lengths in the two triangles. Finally, students discover that the midsegment is half the length of the third side, which it is parallel to.



11.8.2 Exploration: Triangle Midsegment Theorem

$\triangle NKV$ and $\triangle NRT$ are shown, where $\overline{KV} \parallel \overline{RT}$.



- Discuss with your partner what can be concluded from the given information. Justify your conclusions.
Answers will vary. Because the parallel lines create two sets of congruent corresponding angles, the two triangles are similar by AA similarity.
- Ask a classmate for two conclusions they made. Record their conclusions and write your summary of their reason. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

Conclusion	My Summary of Their Reason	Initials
<i>Answers will vary.</i>		
<i>Answers will vary.</i>		



Collect and Display

Listen for, collect, and display student conjectures about the relationship between the triangles. Display the Angle-Angle proof of triangle similarity on a classroom Word Wall or anchor chart.



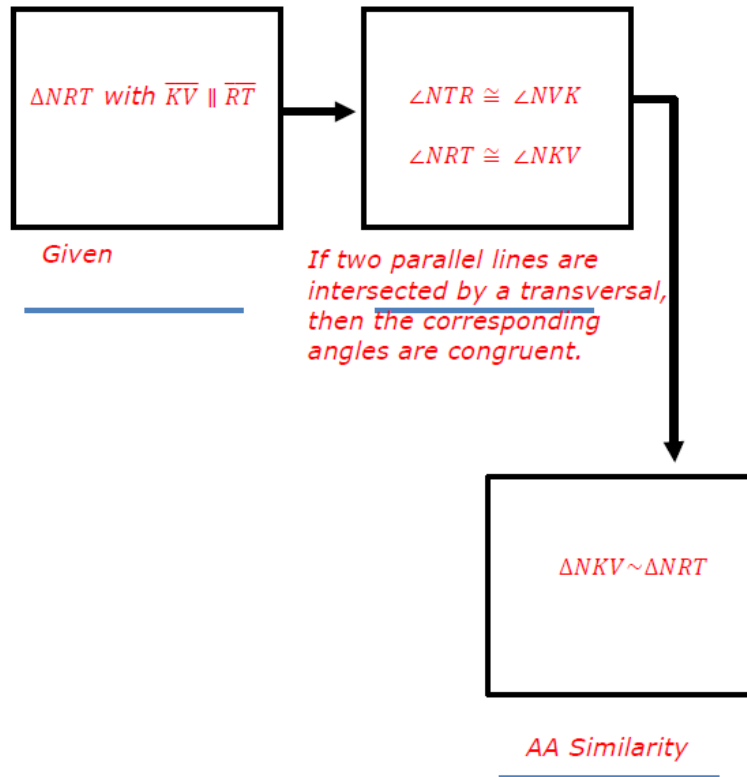
- What are your conclusions?
- Do you agree or disagree with your peers' conclusions? Why or why not?

11.8.2 Exploration: Triangle Midsegment Theorem (continued)

3. Given: $\triangle NRT$ with $\overline{KV} \parallel \overline{RT}$

Prove: $\triangle NKV \sim \triangle NRT$

Complete the flowchart proof.



Provide an auditory entry point by having students discuss what they notice about the relationship between the triangles. The flow chart proof provides students with a visual entry point. It also includes less words than a traditional proof, helping students comprehend the concept.

11.8.2 Exploration: Triangle Midsegment Theorem (continued)

4. What other information about triangles NKV and NRT can be determined from knowing that $\triangle NKV \sim \triangle NRT$?

Answers will vary. All of the corresponding sides will be proportional.

5. Complete the table using the information $KV = 5$, $NR = 6$, $NK = 3$, and $NT = 8.8$.

$RT = 10$	$NV = 4.4$
$KR = 3$	$VT = 4.4$

6. What do you notice about the location of $\overline{KV}$?
It connects the midpoints of $\overline{NR}$ and $\overline{NT}$.
7. What do you notice about the lengths of $\overline{KV}$ and $\overline{RT}$?
 $\overline{KV}$ is half the length of $\overline{RT}$.



Triangle Midsegment Theorem

The midsegment of a triangle joins the midpoints of two sides of a triangle such that its length is half the length of the third side of the triangle, and it is parallel to the third side of the triangle.



Consider having students label the triangle with the information given and found in question 5. This could provide students with a better understanding of the location and lengths by providing students with a visual.



Think-Pair-Share

Consider providing students with the opportunity to share with a partner what they notice after individual think time. Engaging the class in a group discussion will provide entry points for auditory learners.

11.8.3 Guided Practice: Triangle Midsegment Theorem

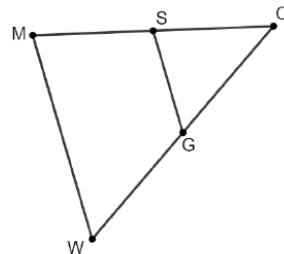
Component Overview: Students use the triangle midsegment theorem to solve multi-step problems. Model how to solve the problems and support students as they work during this component.



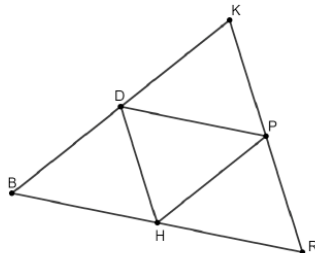
11.8.3 Guided Practice: Triangle Midsegment Theorem

1. $\triangle MCW$ is shown, where $\overline{SG}$ is the midsegment.

$$\begin{aligned} MS &= \left(\frac{1}{2}x + 4\right), \\ GC &= (11z - 2), \\ SG &= (2y + 5), \\ MW &= (8y - 2), \\ SC &= \left(\frac{3}{4}x - \frac{1}{4}\right), \\ WG &= (9z + 1). \end{aligned}$$



- Find the value of x .
 $x = 17$
 - Find the value of y .
 $y = 3$
 - Find the value of z .
 $z = 1.5$
 - What is the perimeter of $\triangle MCW$?
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2. $\triangle BKR$ is shown, where D , P , and H are the midpoints and the perimeter of $\triangle DPH$ is 23.5. $DP = 17x$, $PH = (6x + 5)$, and $HD = 20x - 3$.



- What is the value of x ?
 $x = 0.5$



For this Guided Practice, determine how much guidance is needed for your students based on observations during the 11.8.2 Exploration. If necessary, consider modeling how to solve question 1 to address any misconceptions students may have.



Think-Pair-Share

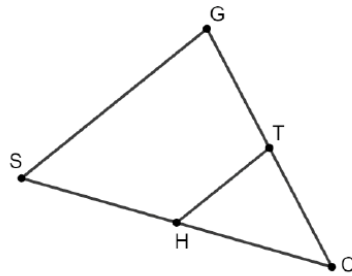
Prompt students to initially try the problems independently. Next, ask students to discuss their strategies and answers with a partner. Finally, call on a variety of students to share with the whole class during a debrief discussion. Ask other students if they agree or disagree and why.

11.8.3 Guided Practice: Triangle Midsegment Theorem (continued)

b. Complete the table.

$BK = 16$	$DH = 7$
$BR = 17$	$KP = 7$
$DP = 8.5$	$KR = 14$

3. $\triangle SGC$ and $\triangle HTC$ are shown.



Given: $SH = HC$ and $GT = TC$

Prove: $\overline{SG} \parallel \overline{HT}$

Since $SH = HC$ and $GT = TC$, point H is the midpoint of $\overline{SC}$ and point T is the midpoint of $\overline{GC}$ by the definition of midpoint. Therefore, $\overline{HT}$ is the midsegment of $\triangle SGC$ by definition and $\overline{SG} \parallel \overline{HT}$ by the triangle midsegment theorem.



Provide a kinesthetic entry point by having students use Geoboards, Magformers, Busy Bags, or Exploragons to physically model the diagram.



- How do you find the lengths of each line segment after finding the value of the variable?
- How can you check your work?



Consider providing students with a bank of words to choose from to complete the paragraph in question 3. Explain that words in the bank can be used more than once. Include words that are not used in the word bank. Prompt students to read and rewrite the paragraph several times to help them internalize the concepts.

11.8.4 Your Turn!

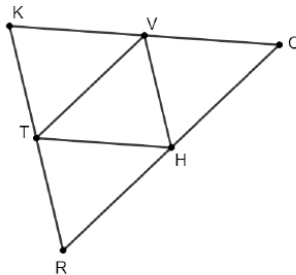
Component Overview: Students use the triangle midsegment theorem to independently solve problems. Circulate while students work, to provide scaffolds to struggling students and determine which student to call on during the debrief discussion.



11.8.4 Your Turn!

$\triangle KCR$ is shown, where T , V , and H are midpoints.

$KC = (5x)$, $VH = (\frac{1}{3}y + 11)$, $VT = (6w - 5)$, $KR = (2y - 2)$,
 $RC = (8w + 8)$, and $TH = (3x - 4)$.



- Find the value of each variable.

$x = 8$	$y = 18$	$w = 4.5$
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- Complete the table.

$TH = 20$	$KC = 40$
$VH = 17$	$KR = 34$
$VT = 22$	$RC = 44$



Clarify, Critique, Correct

Consider using the “Clarify, Critique, Correct” Instructional Routine for this activity. Have partners exchange their work to analyze, reflect on, and develop their partner’s work. This activity will help students improve upon their work by correcting errors and clarifying the meaning of their partner’s work. This routine fortifies output and engages students in meta-awareness.

11.8.5 Wrap-Up: Error Analysis

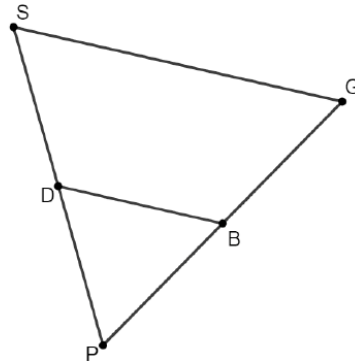
Component Overview: Students identify and correct errors in a sample of student work.



11.8.5 Wrap-Up: Error Analysis

1. $\triangle SGP$ is shown, where $\overline{DB}$ is a midsegment. $SG = 18x + 22$ and $DB = 11x + 7$.

Priscilla was finding the value of x . Her work is shown.



Priscilla's Work

$$\begin{aligned} 18x + 22 &= 11x + 7 \\ 7x &= -15 \\ x &= -\frac{15}{7} \end{aligned}$$

- a. Write a note to Priscilla explaining her mistake.
Priscilla, when you set up your equation, you set the two lengths equal to each other, but the length of $\overline{DB}$ is half the length of $\overline{SG}$.
- b. Find the value of x .
 $x = 2$



Students have the opportunity to demonstrate their mastery of the learning targets by identifying and correcting errors in a sample of student work. Discussing and reflecting on the sample student errors will decrease the likelihood of students making similar errors themselves in the future.